Torque Ripple and Cogging Torque in Permanent-Magnet Synchronous Machines

Abstract

This note describes the physical origin of torque ripple in permanent-magnet synchronous machines (PMSM), separates the current-independent contribution (cogging torque) from the current-dependent contributions (back-EMF harmonics, reluctance modulation, inverter and sensor imperfections), and derives the harmonic orders associated with each mechanism. A dedicated section analyses how the uncompensated ripple propagates through a back-to-back AFE + inverter structure with a reduced DC-link capacitance, and quantifies the resulting interharmonic emission on a 50 Hz grid for a machine running at a fundamental frequency of 15.6 Hz. Design-side, control-side and grid-side mitigation methods are summarised at the end.

1 Torque production in the ideal machine

In the ideal PMSM the phase flux linkages produced by the magnets are purely sinusoidal functions of rotor position, the airgap permeance is constant, and the machine is fed with sinusoidal balanced currents. Under these assumptions the torque in the rotor reference frame is

$$T_e = \frac{3}{2}\, p \left[\lambda_{pm}\, i_q + (L_d - L_q)\, i_d i_q\right], \tag{1}$$

where p is the number of pole pairs, λ_{pm} the magnet flux linkage (peak, per phase), and L_d, L_q the synchronous inductances. Both terms are constant for constant i_d, i_q: the ideal machine produces no ripple.

Every ripple mechanism discussed below is therefore a violation of one of the three assumptions: sinusoidal flux-linkage distribution, position-independent permeance, or ideal current control.

2 Energy-based formulation

The complete instantaneous torque follows from the derivative of the magnetic co-energy $W'(\mathbf{i}, \theta_m)$ at constant current:

$$T_e(\mathbf{i}, \theta_m) = \left.\frac{\partial W'(\mathbf{i}, \theta_m)}{\partial \theta_m}\right|_{\mathbf{i}=\text{const}}. \tag{2}$$

Writing W' as the sum of a winding term, a mutual (magnet–winding) term and a magnet-only term gives the standard decomposition

$$T_e = \underbrace{\mathbf{i}^{\mathsf{T}} \frac{\partial \boldsymbol{\lambda}_{pm}(\theta_m)}{\partial \theta_m}}_{\text{alignment (mutual) torque}} + \underbrace{\frac{1}{2}\, \mathbf{i}^{\mathsf{T}} \frac{\partial \mathbf{L}(\theta_m, \mathbf{i})}{\partial \theta_m}\, \mathbf{i}}_{\text{reluctance torque}} + \underbrace{\frac{\partial W'_{pm}(\theta_m)}{\partial \theta_m}}_{\text{cogging torque}}. \tag{3}$$

The three terms have fundamentally different behaviour:

- the first is proportional to current and reproduces the harmonic content of $\partial \boldsymbol{\lambda}_{pm}/\partial \theta_m$, i.e. of the back-EMF;

- the second is proportional to the square of the current and depends on the position (and current) dependence of the inductance matrix, i.e. on saliency, slotting and saturation;
- the third is independent of the current and exists even with the machine unexcited. This is the cogging torque.

3 Cogging torque

3.1 Physical origin

Cogging torque is the tendency of the rotor to align itself in positions of minimum reluctance for the magnet flux. It exists because the stator is slotted: as the rotor turns, the magnets see an airgap permeance that is modulated by the alternation of teeth and slot openings, so the stored magnetic energy of the magnet field varies with position. Any variation of energy with position is a torque.

A first-order description uses the product of the magnet magnetomotive force and the slotted-airgap permeance function referred to the stator:

$$F(\theta, \theta_m) = \sum_{m} F_m \cos\big[mp(\theta - \theta_m)\big], \qquad \Lambda(\theta) = \Lambda_0 + \sum_{n \geq 1} \Lambda_n \cos\big(nQ\,\theta\big), \tag{4}$$

with Q the number of stator slots. The airgap energy contains products $F_m \Lambda_n$; after integration over the airgap circumference, only the terms satisfying

$$nQ = m\,p \tag{5}$$

survive and depend on θ_m. The resulting torque is periodic with

$$N_c = \mathrm{lcm}(Q,\, 2p) \tag{6}$$

periods per mechanical revolution, and can be written as a Fourier series

$$T_{cog}(\theta_m) = \sum_{k \geq 1} T_{cog,k}\, \sin\big(kN_c\theta_m + \varphi_k\big). \tag{7}$$

The series has zero mean over one revolution: cogging torque does not contribute average torque, only ripple.

Referred to the electrical period, the fundamental cogging order is N_c/p. For $Q = 12$, $2p = 10$ ($p = 5$): $N_c = 60$ mechanical periods per revolution, i.e. a 12th electrical harmonic.

3.2 Design factors

The amplitude $T_{cog,1}$ decreases as N_c increases, and N_c is maximised when $\gcd(Q, 2p)$ is small; $\gcd(Q, 2p)$ is therefore commonly used as a cogging "goodness" indicator (lower is better). Other dominant factors are the slot-opening-to-airgap ratio, the pole-arc to pole-pitch ratio, the magnetisation profile, and skewing.

Continuous skew over an angle θ_{sk} attenuates the k-th cogging harmonic by

$$k_{sk,k} = \frac{\sin\big(kN_c\theta_{sk}/2\big)}{kN_c\theta_{sk}/2}, \tag{8}$$

which vanishes for $\theta_{sk} = 2\pi/N_c$, i.e. one full cogging period. Step skew of the rotor magnets in n_s axial blocks gives a comparable attenuation with simpler manufacturing.

4 Current-dependent torque ripple

4.1 Back-EMF harmonics

Real magnet flux-linkage distributions are not sinusoidal. Writing the phase flux linkage as

$$\lambda_a(\theta_e) = \sum_{h\ \text{odd}} \lambda_h \cos(h\theta_e), \tag{9}$$

the corresponding back-EMF is $e_a = \omega_e \sum_h h\lambda_h \sin(h\theta_e)$, i.e. the harmonic content is amplified by the harmonic order h.

Feeding the machine with a purely sinusoidal current $i_a = I\sin(\theta_e)$ (and the corresponding $\pm 120°$ phase shifts), the instantaneous torque is obtained from the power balance $T_e = \big(e_a i_a + e_b i_b + e_c i_c\big)/\omega_m$. Each product $\sin(h\theta_e)\sin(\theta_e)$ generates components of order $h-1$ and $h+1$; summing the three phases cancels every order that is not a multiple of six, and the triplen flux-linkage harmonics produce only zero-sequence EMF, which develops no torque in a star-connected machine without neutral. The result is

$$T_e(\theta_e) = \frac{3}{2}\,p\,I\left[\lambda_1 + \sum_{n\geq 1}\Big((6n{+}1)\lambda_{6n+1} - (6n{-}1)\lambda_{6n-1}\Big)\cos(6n\theta_e)\right]. \tag{10}$$

Equation (10) is the central result for this ripple family: the 5th and 7th back-EMF harmonics produce a 6th-order torque ripple, the 11th and 13th produce a 12th-order ripple, and so on. Note the sign: the two contributions may partially cancel, so a machine with a distorted back-EMF is not necessarily a machine with large ripple.

The dual statement also holds. If the back-EMF is sinusoidal but the current contains 5th and 7th harmonics, the same 6th-order ripple appears. In the rotor frame both cases appear as a $6k$ ripple superimposed on i_d, i_q and on the torque.

4.2 Reluctance and saturation effects

In interior-magnet machines the inductances are position- and current-dependent because of slotting, rotor geometry and local saturation of the bridges and tooth tips. Expanding $L_d(i_d, i_q, \theta_e)$ and $L_q(i_d, i_q, \theta_e)$ in (3) yields additional $6k$ terms whose amplitude grows with the square of the current, so this contribution dominates at high load while cogging dominates at light load. Cross-saturation ($\partial\lambda_d/\partial i_q \neq 0$) adds a further load-dependent component and detunes the MTPA angle.

4.3 Inverter and control-chain contributions

These sources are independent of the machine design and are frequently the dominant ones in a well-designed motor:

- **Dead time and device voltage drop.** The distortion of the applied voltage is a square-wave function of the current sign, generating $6k \pm 1$ current harmonics and therefore $6k$ torque ripple. The effect is largest at low speed and low current, where the distortion voltage is comparable with the fundamental.
- **Current-sensor DC offset.** An offset on one phase appears as a rotating current component synchronous with the rotor, producing a 1st-order torque ripple.
- **Current-sensor gain mismatch.** Unequal scaling between phases produces a negative-sequence fundamental component and hence a 2nd-order torque ripple.

- **Position-sensor error.** Resolver eccentricity or amplitude imbalance produces 1st- and 2nd-order errors in the estimated angle, which modulate the torque angle and thus the torque. Encoder quantisation and observer lag act similarly at higher orders.
- **DC-link voltage ripple.** A ripple at $2f_{grid}$ (single-phase front end) or $6f_{grid}$ (six-pulse rectifier) propagates through the modulator into the currents; in voltage saturation it appears directly as torque ripple at the same frequency, asynchronous with the rotor.
- **Finite current-loop bandwidth.** The current controller can only reject disturbances well below its crossover. At high speed $6f_e$ may exceed the achievable bandwidth, so the ripple grows with speed even when the machine itself is unchanged.

4.4 Manufacturing asymmetries

Static or dynamic rotor eccentricity, magnet dimensional and remanence tolerances, and partial demagnetisation break the symmetry between poles and introduce sub-harmonics, typically 1st- and 2nd-order components, together with unbalanced magnetic pull and additional acoustic noise.

5 Summary of harmonic orders

Source	Order (electrical)	Current dependence
Cogging	$k\,N_c/p, \quad N_c = \mathrm{lcm}(Q, 2p)$	none
Back-EMF harmonics $6n \pm 1$	$6n$	linear
Current harmonics $6n \pm 1$	$6n$	linear
Saliency, slotting inductance modulation	$6n$	quadratic
Saturation and cross-saturation	$6n$	quadratic, load dependent
Dead time	$6n$	strongest at low current
Sensor offset	1	constant absolute value
Sensor gain mismatch	2	linear
Position error (resolver)	1, 2	linear
Rotor eccentricity, magnet tolerance	1, 2	linear
DC-link ripple	asynchronous ($2f_g$, $6f_g$)	linear

Table 1: Dominant torque-ripple mechanisms and their harmonic signature.

A practical consequence: the harmonic order observed in a torque or speed spectrum identifies the mechanism. A pure 6th order that scales with current points to back-EMF or current distortion; a 6th order that scales with current squared points to saturation; a component present at zero current is cogging; 1st- and 2nd-order components point to the sensing chain or to mechanical asymmetry.

6 Propagation into the DC link and into the grid

This section analyses the following scenario: a back-to-back structure with an active front end (AFE) and a motor-side inverter sharing a *light* DC link (film capacitor only, no electrolytic bank, no DC choke), feeding a PMSM at a fundamental frequency $f_e = 15.6\,\text{Hz}$ on a $f_g = 50\,\text{Hz}$ grid, with *no* torque-ripple compensation on the motor side. The ripple mechanisms of Sections 3–4 are therefore present at full amplitude, and the question is where the associated pulsating power ends up.

6.1 Instantaneous power balance

Neglecting losses and the rate of change of stored magnetic energy, the power delivered by the motor-side inverter equals the shaft power. A torque ripple component of order n therefore corresponds to a pulsating power

$$\tilde{p}_n(t) = \tilde{T}_n\,\omega_m \cos(n\omega_e t + \varphi_n), \qquad \omega_m = \frac{\omega_e}{p}, \tag{11}$$

so that, in per-unit terms, the pulsating power fraction equals the torque ripple fraction:

$$\frac{\hat{\tilde{p}}_n}{P_{sh}} = \frac{\tilde{T}_n}{T}. \tag{12}$$

Cogging is an exception. Cogging torque is not supplied electrically: it is an energy exchange between the magnet field and the rotor inertia. It reaches the DC link only indirectly, through the speed ripple it produces, $\Delta\hat{\omega}_m \approx \tilde{T}_{cog}/(J\,n\omega_e)$, and through the reaction of the speed and current loops. At $n\omega_e$ well above the mechanical bandwidth this path is strongly attenuated by the inertia. The DC-side pulsation is therefore dominated by the *current-dependent* ripple: back-EMF harmonics, saturation-induced inductance modulation, dead time and sensor errors.

6.2 Split between capacitor and grid

The pulsating power (11) can only be absorbed by the DC-link capacitor or transferred to the grid through the AFE. The split is set by the DC-voltage control loop. Denoting by $T_v(s)$ its complementary sensitivity (closed-loop transfer from power disturbance to grid-side power),

$$\tilde{p}_{grid} = |T_v(jn\omega_e)|\,\tilde{p}_n, \qquad \tilde{p}_C = |1 - T_v(jn\omega_e)|\,\tilde{p}_n, \tag{13}$$

and the resulting DC-link voltage ripple is

$$\Delta\hat{V}_{dc,n} = \frac{|1 - T_v(jn\omega_e)|\,\hat{\tilde{p}}_n}{n\omega_e\,C_{dc}\,V_{dc}} = \frac{|1 - T_v(jn\omega_e)|\,\tilde{T}_n}{n\,p\,C_{dc}\,V_{dc}}. \tag{14}$$

Equation (14) contains the first non-obvious result: with the DC-voltage loop inactive at the ripple frequency, the DC-link voltage ripple caused by torque ripple is **independent of speed**. The pulsating power grows linearly with ω_e, while the capacitor impedance falls as $1/(n\omega_e)$, and the two effects cancel exactly. Operating at low fundamental frequency does not reduce the DC-link voltage ripple; it only reduces the pulsating power in absolute terms.

The second result concerns the split itself. A DC-voltage loop with closed-loop bandwidth f_{bw} and a single dominant pole rejects the disturbance with $|T_v| \approx f_{bw}/(nf_e)$ above crossover. At $f_e = 50\,\text{Hz}$ the sixth-order ripple sits at 300 Hz and only a few percent reaches the grid; at $f_e = 15.6\,\text{Hz}$ it sits at 93.6 Hz, i.e. only about three times the typical DC-loop bandwidth, and roughly a third of the pulsating power is transferred to the grid. **Low-speed operation is the worst case for grid-side emission**, even though the absolute pulsating power is smaller.

An equivalent energy view is useful for sizing: the energy that must swing in the capacitor over one ripple period is $\Delta W = \hat{\tilde{p}}_n/(n\omega_e)$, to be compared with the stored energy $W_C = \frac{1}{2}C_{dc}V_{dc}^2$, giving $\Delta\hat{V}_{dc}/V_{dc} \approx \Delta W/(2W_C)$.

6.3 Worked example

Table 2 collects a representative case. The torque ripple is assumed to be 3 % of rated torque at the sixth order, which is a realistic figure for an uncompensated drive with dead time and a moderately distorted back-EMF.

Quantity	Symbol	Value
Grid	V_g, f_g	690 V, 50 Hz
Short-circuit ratio	S_{sc}/S_n	20
Rated power	P_n	250 kW
DC-link voltage	V_{dc}	1100 V
DC-link capacitance	C_{dc}	200 µF ($\approx 0.8\,\mu\mathrm{F\,kW^{-1}}$)
Pole pairs	p	4
Fundamental frequency	f_e	15.6 Hz ($n = 234$ rpm)
Rated torque	T_n	10.2 kN m
Torque ripple, 6th order	$\tilde{T}_6$	3 % = 306 N m
Ripple frequency	$6f_e$	93.6 Hz
Pulsating power (peak)	$\hat{\tilde{p}}_6$	7.5 kW
DC current ripple (peak)	$\hat{\tilde{i}}_{dc}$	6.8 A
DC-loop bandwidth	f_{bw}	30 Hz $\Rightarrow |T_v| \approx 0.32$
Power to the grid	$\tilde{p}_{grid}$	2.4 kW
DC-link voltage ripple	$\Delta\hat{V}_{dc}$	39 V = 3.6 %
Grid fundamental current	I_1	209 A
Each interharmonic sideband	I_{ih}	1.0 A = 0.48 % of I_1

Table 2: Representative uncompensated case, AFE + inverter with light DC link.

The sideband amplitude follows from the pulsating power on the grid side. With amplitude-invariant Park and unity displacement factor, $\hat{\tilde{p}}_{grid} = \frac{3}{2}\hat{V}_{ph}\,\hat{i}_{d,r}$, so

$$\hat{i}_{d,r} = \frac{2\,\hat{\tilde{p}}_{grid}}{3\,\hat{V}_{ph}}, \tag{15}$$

and a d-axis component oscillating at $\omega_r = n\omega_e$ maps into a *pair* of phase-current components of amplitude $\hat{i}_{d,r}/2$ each.

6.4 Interharmonic sidebands

Because the DC-link ripple modulates the AFE, every DC-side ripple frequency f_r appears in the grid current as a pair of sidebands

$$f_{ih}^{\pm} = |f_g \pm f_r| = |f_g \pm n f_e|. \tag{16}$$

Since f_e and f_g are unrelated, these are *interharmonics*, not harmonics. When $f_g - f_r < 0$ the lower component appears with reversed phase sequence, i.e. as a negative-sequence interharmonic. Table 3 applies (16) to the case at hand.

Ripple order n	Origin	f_r [Hz]	Sidebands [Hz]	Beat with f_g [Hz]
1	sensor offset, eccentricity	15.6	34.4, 65.6	15.6
2	sensor gain mismatch	31.2	18.8, 81.2	31.2
6	back-EMF, dead time, sat.	93.6	**43.6** (neg. seq.), 143.6	**6.4**
12	back-EMF, cogging (typ.)	187.2	137.2, 237.2	87.2
18	higher orders	280.8	230.8, 330.8	180.8

Table 3: Interharmonic emission produced by an uncompensated drive at $f_e = 15.6$ Hz on a 50 Hz grid.

6.5 Flicker mechanism: the $|2f_g - 6f_e|$ beat

The interharmonic pair of Table 3 does not merely sit near the fundamental: it produces a genuine low-frequency fluctuation of the per-phase power, and therefore of the per-phase voltage

drop across the supply impedance. Three equivalent derivations lead to the same beat frequency.

(i) Envelope view. The lower sideband at $f_{ih}^{-} = |f_g - 6f_e| = 43.6\,\text{Hz}$ superimposes on the 50 Hz fundamental; in each phase the two components beat, and the envelope of the phase voltage fluctuates at $|f_g - f_{ih}^{-}| = 6.4\,\text{Hz}$.

(ii) Power view. Taking one phase, with $v_a = \hat{V}\cos(\omega_g t)$ and the interharmonic current $i_{ih} = \hat{I}_{ih}\cos(\omega_{ih}t + \phi)$, the instantaneous single-phase power is

$$p_a(t) = \frac{\hat{V}\hat{I}_{ih}}{2}\Big[\cos\big((\omega_g - \omega_{ih})t - \phi\big) + \cos\big((\omega_g + \omega_{ih})t + \phi\big)\Big]. \tag{17}$$

Substituting $\omega_{ih} = 6\omega_e - \omega_g$ gives the two terms at

$$|2f_g - 6f_e| = 6.4\,\text{Hz} \qquad \text{and} \qquad 6f_e = 93.6\,\text{Hz}, \tag{18}$$

which is exactly the $100\,\text{Hz} - 93.6\,\text{Hz}$ arithmetic. The two views are algebraically identical:

$$f_{beat} = \Big|f_g - \big|f_g - nf_e\big|\Big| = \big|2f_g - nf_e\big| \quad \text{for } nf_e > f_g. \tag{19}$$

The phase of (17) determines how the fluctuation splits between active and reactive power. Since the resulting voltage fluctuation at the PCC is

$$\frac{\Delta V}{V} \approx \frac{R\,\Delta P + X\,\Delta Q}{V^2}, \tag{20}$$

on MV networks with $X/R > 3$ it is the *reactive* fluctuation that dominates the voltage modulation, and hence the flicker.

(iii) Flickermeter view. Block 2 of the IEC 61000-4-15 flickermeter squares the input voltage. The square of the sum of a 50 Hz fundamental and a 43.6 Hz interharmonic contains a component at 6.4 Hz and one at 93.6 Hz; the weighting filter chain, a bandpass of roughly 0.05 Hz to 35 Hz followed by the eye–lamp weighting, passes the first and rejects the second. The instrument therefore responds to precisely the term identified in (17).

Why a three-phase measurement hides it. The lower sideband is negative-sequence, so the 6.4 Hz terms of (17) appear in the three phases displaced by $\mp 120°$ and cancel in the three-phase sum: the total instantaneous power, and equivalently the space-vector magnitude or any *dq*-domain quantity, oscillates only at 93.6 Hz. The voltage drop across the supply impedance is however a per-phase quantity, and lamps are single-phase loads. A *dq*-based analysis of the AFE will therefore never reveal this flicker mechanism; it must be evaluated per phase.

Critical fundamental frequencies. Applying (19) with $n = 6$, and taking the flicker-relevant band as roughly 0.5 Hz to 25 Hz with maximum sensitivity near 8.8 Hz, the sixth-order torque ripple generates flicker whenever

$$\big|2f_g - 6f_e\big| < 25\,\text{Hz} \qquad \Longrightarrow \qquad 12.5\,\text{Hz} < f_e < 20.8\,\text{Hz} \quad (f_g = 50\,\text{Hz}), \tag{21}$$

with the worst case at $f_e = (2f_g - 8.8)/6 = 15.2\,\text{Hz}$. Table 4 maps the band.

The operating point of 15.6 Hz lies 0.4 Hz from the worst possible fundamental frequency for flicker generated by the sixth-order torque ripple. This is a coincidence of the application, not a property of the machine, and it is the single most important conclusion of this section: the same drive at 25 Hz ($f_{beat} = 50\,\text{Hz}$) or at 10 Hz ($f_{beat} = 40\,\text{Hz}$) would produce the same DC-link ripple and the same interharmonic current, with no flicker at all.

f_e [Hz]	$6f_e$ [Hz]	f_{ih}^- [Hz]	f_{beat} [Hz]	Remark
12.5	75.0	25.0	25.0	lower edge of the flicker band
14.0	84.0	34.0	16.0	inside the band
15.2	91.2	41.2	8.8	**maximum flicker sensitivity**
15.6	93.6	43.6	6.4	**present operating point**
16.67	100.0	50.0	0	steady negative sequence: unbalance, not flicker
18.0	108.0	58.0	8.0	inside the band
20.8	124.8	74.8	24.8	upper edge of the flicker band

Table 4: Beat frequency produced by the sixth-order torque ripple as a function of the motor fundamental frequency, $f_g = 50\,\text{Hz}$.

Second-order mechanism: intermodulation with a real 100 Hz ripple. If the grid voltage is unbalanced, or the AFE current sensing is unbalanced, the front end already produces a genuine power and DC-link ripple at $2f_g = 100\,\text{Hz}$. This component coexists with the 93.6 Hz component from the motor. The superposition is linear in the capacitor voltage, but the modulators multiply their duty cycles by V_{dc}, so the two ripples *intermodulate*: a component at $|2f_g - 6f_e| = 6.4\,\text{Hz}$ appears directly in the applied voltages, producing a *symmetric* sideband pair at $f_g \pm 6.4\,\text{Hz} = 43.6$ and $56.4\,\text{Hz}$. A symmetric pair is pure amplitude modulation of the fundamental and is the most efficient possible flicker source. Its amplitude is proportional to the *product* of the two relative DC-link ripples, hence second order and normally smaller than the mechanism above, but it grows as $1/C_{dc}^2$ and therefore becomes relevant exactly in the light-DC-link case considered here. A fast, unfiltered V_{dc} feedforward in both modulators removes this path; a slow or heavily filtered one leaves it open.

6.6 Why 15.6 Hz is an unfavourable operating point

Three independent reasons make this particular fundamental frequency critical.

(a) The beat falls in the flicker band. As shown in Section 6.5, the sixth-order ripple produces a per-phase power and voltage fluctuation at $|2f_g - 6f_e| = 6.4\,\text{Hz}$, essentially at the minimum of the IEC 61000-4-15 perceptibility curve. Writing $a = V_{ih}/V_1$, the relative voltage fluctuation seen by the flickermeter is $\Delta V/V \approx 2a$, and

$$a \approx \frac{I_{ih}}{I_n} \cdot \frac{S_n}{S_{sc}} \cdot \frac{f_{ih}}{f_g} \tag{22}$$

for a predominantly inductive grid impedance. For Table 2: $a \approx 0.021\,\%$, hence $\Delta V/V \approx 0.042\,\%$ and, against a threshold of roughly $0.33\,\%$ for $P_{st} = 1$ at that beat frequency, P_{st} of the order of 0.1. The scaling law is what matters in practice:

$$P_{st} \;\propto\; \frac{\tilde{T}_6}{T_n} \cdot \frac{S_n}{S_{sc}}, \tag{23}$$

so the same drive on a weak grid ($S_{sc}/S_n = 5$) with $6\,\%$ torque ripple reaches $P_{st} \approx 1$, i.e. the planning level. Note that this mechanism is absent on a 60 Hz grid at the same speed, where the lower sideband would fall at 33.6 Hz and beat at 26.4 Hz, well outside the sensitive band.

(b) Proximity to the unbalance-rejection frequency. At $f_e = f_g/3 = 16.67\,\text{Hz}$ the sixth-order ripple would sit exactly at $2f_g = 100\,\text{Hz}$, and the lower sideband would land exactly on 50 Hz with negative sequence, i.e. it would be indistinguishable from grid voltage unbalance and would be actively fought by the AFE negative-sequence controller. The operating point of 15.6 Hz is only 1.1 Hz away from this singular condition, so the emission falls in the transition band of the notch filters or DSC/DDSRF prefilters that are usually tuned at 100 Hz: they neither pass

it cleanly nor reject it, and the resulting partial, phase-shifted response can degrade the AFE current-loop damping. A second singular point is $f_e = f_g/6 = 8.33\,\text{Hz}$, where the lower sideband collapses to DC and the drive injects a DC current component into the grid, with the associated transformer-saturation risk.

(c) Low ripple frequency relative to the DC-voltage loop. As shown above, at 93.6 Hz the DC-voltage loop is not yet fully rolled off, so a substantial fraction of the pulsating power is deliberately pushed onto the grid by the controller itself.

6.7 Secondary effects of a light DC link

- **Loss of voltage headroom.** A 3.6 % DC-link ripple reduces the voltage available to the motor-side modulator at every trough. If the drive operates near the modulation limit, the inverter enters overmodulation periodically at 93.6 Hz, generating additional low-order current harmonics which in turn increase the torque ripple: a positive-feedback path that does not exist with a stiff DC link.
- **Direct modulation path.** Even with a very slow DC-voltage loop, the ripple reaches the grid current through the modulator: both converters normalise their duty cycles by V_{dc}, and any delay or filtering in that feedforward leaves a residual voltage error at 93.6 Hz which the current loops must reject. This path is independent of T_v and becomes dominant as C_{dc} is reduced.
- **Constant-power-load interaction.** At frequencies below its bandwidth the motor drive behaves as a constant-power load with negative incremental resistance $-V_{dc}^2/P$. With a light DC link the AFE closed-loop output impedance is larger and less damped in the 50 Hz to 200 Hz region, exactly where the 93.6 Hz excitation lies; the ripple can therefore be amplified rather than attenuated, and the impedance-ratio stability margin should be verified at that frequency rather than assumed.
- **Capacitor stress.** The RMS current in a film capacitor of 200 µF is dominated by the switching-frequency component, but the 93.6 Hz component adds a low-frequency thermal cycle; more importantly, the voltage ripple must be included in the derating and lifetime calculation.
- **Measurement.** IEC 61000-4-7 uses 200 ms windows with 5 Hz resolution. The 43.6 Hz component falls between spectral lines and close to the fundamental, so leakage from the 50 Hz component can mask it; interharmonic subgroup evaluation and windowing must be used, otherwise the emission is simply not seen in a standard harmonic report.

6.8 Summary of the propagation chain

$$\tilde{T}_6 \;\rightarrow\; \tilde{p} = \tilde{T}_6\omega_m \;\rightarrow\; \Big[\text{capacitor: } \Delta V_{dc}\Big] \;\Big\|\; \Big[\text{AFE: } \tilde{p}_{grid}\Big] \;\rightarrow\; f_g \pm 6f_e \;\rightarrow\; \text{flicker, interharmonics}$$

With a light DC link and no motor-side compensation there is no way to make the pulsation disappear: the design choice is only *where to put it*. Either it is absorbed by the capacitor, and then the DC-link voltage ripple and the associated headroom loss must be budgeted, or it is passed to the grid, and then the interharmonic emission and flicker must be assessed against IEC 61000-2-2 / IEC 61000-3-x compatibility levels, EN 61800-3 and, where applicable, IEEE 519. Reducing the ripple at the source on the motor side is the only option that avoids both.

7 Mitigation

7.1 Machine design

- Slot/pole combination with large $\mathrm{lcm}(Q, 2p)$ and small $\gcd(Q, 2p)$.
- Continuous or step skewing of stator slots or rotor magnets.
- Magnet shaping: pole-arc ratio optimisation, eccentric or sinusoidally profiled magnet surface, magnet segmentation and pole shifting.
- Semi-closed slots, reduced slot opening, magnetic slot wedges, dummy slots or tooth notches to increase the effective slotting frequency.
- Winding layout (fractional-slot concentrated versus distributed) chosen considering that it changes both the back-EMF harmonic content and the cogging periodicity.

Skewing reduces cogging and back-EMF harmonics but slightly reduces the fundamental torque and complicates manufacturing; the design is a compromise between ripple, torque density and cost.

7.2 Control

Control-side compensation acts on the current reference or on the voltage command:

- **Harmonic current injection (feedforward).** The reference is augmented with the harmonics that cancel the measured ripple:

$$i_q^*(\theta_e) = i_{q0}^* + \sum_n A_n \cos(6n\theta_e + \phi_n), \tag{24}$$

  with A_n, ϕ_n identified offline and stored in a speed/load-indexed lookup table.

- **Resonant (PR / vector resonant) controllers** added in parallel to the synchronous-frame PI at the ripple frequencies,

$$G_r(s) = \frac{2k_r\,\omega_c\,s}{s^2 + 2\omega_c s + (6n\omega_e)^2}, \tag{25}$$

  providing high loop gain at $6n\omega_e$ without affecting the DC behaviour.

- **Repetitive control** in the position domain, whose internal model covers all harmonics of the mechanical or electrical period at once; it is the natural choice when the disturbance is position-periodic (cogging) rather than time-periodic.
- **Iterative learning control** for repetitive operating cycles.
- **Disturbance observers** estimating the periodic load torque from the speed signal, with the estimate fed forward to the current reference.
- **Dead-time and voltage-error compensation**, plus current-sensor offset and gain self-calibration, which remove the 1st, 2nd and part of the 6th order without any additional dynamics.

Two limits apply to all active methods. First, they require loop bandwidth at $6n\omega_e$, which becomes the binding constraint at high speed; multiple synchronous frames or position-domain formulations relieve, but do not remove, this constraint. Second, cancelling torque ripple means deliberately injecting current harmonics: this costs additional copper and iron losses and voltage headroom, and the benefit must be weighed against those costs, particularly in the flux-weakening region.

7.3 Grid-side and DC-link measures

When the motor-side ripple cannot be removed, the emission can still be shaped:

- **Notch in the DC-voltage loop** at $6f_e$ (and $12f_e$), tuned adaptively on the measured motor frequency. The AFE then presents a high impedance at the ripple frequency and stops re-exporting the pulsation to the grid; the price is a larger DC-link voltage ripple, which must fit in the modulation headroom of both converters.
- **Deliberate capacitor sizing.** From (14), the capacitance needed for a given ripple budget is $C_{dc} \geq \tilde{T}_n/(n\,p\,V_{dc}\,\Delta\hat{V}_{dc})$, which is independent of speed and therefore a hard sizing constraint rather than a worst-speed calculation.
- **Active power decoupling** on a small auxiliary branch, which buffers the $6f_e$ energy without oversizing the main DC-link capacitor.
- **Speed operating-point selection.** From (19), keeping f_e outside 12.5 Hz to 20.8 Hz (for $f_g = 50\,\text{Hz}$) moves the sixth-order beat out of the flicker band entirely, and avoiding f_e near $f_g/6$ and $f_g/3$ removes the DC-injection and apparent-unbalance conditions. Where the process allows a different gear ratio or a different pole count, this is by far the cheapest measure: it costs nothing and removes the dominant mechanism rather than attenuating it.
- **DC-link voltage feedforward** with adequate bandwidth in both modulators, so that the ripple does not translate into an applied-voltage error.
- **Impedance-based stability check** of the AFE output impedance against the motor-drive input impedance in the 50 Hz to 200 Hz range, which is the frequency band excited by the torque ripple at this operating point.